```
mysql> -- CSA05 - DATABASE MANAGEMENT SYSTEMS
Query OK, 0 rows affected (0.004 sec)

mysql> -- ASSIGNMENT 3 - CO3
Query OK, 0 rows affected (0.003 sec)

mysql> -- SHOPNEST ONLINE RETAIL ORDER MANAGEMENT DATABASE
Query OK, 0 rows affected (0.004 sec)

mysql> -- MYSQL 9.7 COMMAND LINE CLIENT
Query OK, 0 rows affected (0.003 sec)

mysql> -- ========================================================
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> CREATE DATABASE IF NOT EXISTS SHOPNEST_CO3;
Query OK, 1 row affected (0.301 sec)

mysql> USE SHOPNEST_CO3;
Database changed
mysql>
mysql> -- ========================================================
Query OK, 0 rows affected (0.004 sec)

mysql> -- 1. UNNORMALISED TABLE (UNF)
Query OK, 0 rows affected (0.006 sec)

mysql> -- ========================================================
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> DROP TABLE IF EXISTS UNF_ORDER_LINE;
Query OK, 0 rows affected, 1 warning (0.048 sec)

mysql>
mysql> CREATE TABLE UNF_ORDER_LINE (
    ->     OrderID INT NOT NULL,
    ->     OrderDate DATE NOT NULL,
    ->     CustomerID INT NOT NULL,
    ->     CustomerName VARCHAR(100) NOT NULL,
    ->     CustomerEmail VARCHAR(120) NOT NULL,
```

```
Query OK, 0 rows affected (0.387 sec)

mysql>
mysql> INSERT INTO UNF_ORDER_LINE VALUES
    -> (101,'2026-01-10',1,'Arun','arun@gmail.com','Anna Nagar','Chennai','600040','Tamil Nadu',201,'Laptop',55000,301,'Electronics',401,'Tech Supplies','9876543210',1,501,'Chennai','computer,electronics,work','Black,Silver'),
    -> (101,'2026-01-10',1,'Arun','arun@gmail.com','Anna Nagar','Chennai','600040','Tamil Nadu',202,'Mouse',800,301,'Electronics',401,'Tech Supplies','9876543210',2,501,'Chennai','computer,accessory','Black,White'),
    -> (102,'2026-01-11',2,'Bala','bala@gmail.com','Velachery','Chennai','600042','Tamil Nadu',203,'Chair',2500,302,'Furniture',402,'Home Products','9876543211',1,502,'Coimbatore','furniture,office','Brown,Black'),
    -> (102,'2026-01-11',2,'Bala','bala@gmail.com','Velachery','Chennai','600042','Tamil Nadu',204,'Table',4500,302,'Furniture',402,'Home Products','9876543211',1,502,'Coimbatore','furniture,office','Brown,White'),
    -> (103,'2026-01-12',3,'Charan','charan@gmail.com','T Nagar','Chennai','600017','Tamil Nadu',201,'Laptop',55000,301,'Electronics',401,'Tech Supplies','9876543210',1,501,'Chennai','computer,electronics,work','Black,Silver'),
    -> (103,'2026-01-12',3,'Charan','charan@gmail.com','T Nagar','Chennai','600017','Tamil Nadu',205,'Keyboard',1500,301,'Electronics',403,'Digital World','9876543212',1,503,'Madurai','computer,accessory','Black,White'),
    -> (104,'2026-01-13',4,'Deepa','deepa@gmail.com','Gandhipuram','Coimbatore','641012','Tamil Nadu',203,'Chair',2500,302,'Furniture',402,'Home Products','9876543211',2,502,'Coimbatore','furniture,office','Brown,Black'),
    -> (104,'2026-01-13',4,'Deepa','deepa@gmail.com','Gandhipuram','Coimbatore','641012','Tamil Nadu',202,'Mouse',800,301,'Electronics',401,'Tech Supplies','9876543210',1,501,'Chennai','computer,accessory','Black,White'),
    -> (105,'2026-01-14',1,'Arun','arun@gmail.com','Anna Nagar','Chennai','600040','Tamil Nadu',204,'Table',4500,302,'Furniture',402,'Home Products','9876543211',1,502,'Coimbatore','furniture,office','Brown,White'),
    -> (105,'2026-01-14',1,'Arun','arun@gmail.com','Anna Nagar','Chennai','600040','Tamil Nadu',205,'Keyboard',1500,301,'Electronics',403,'Digital World','9876543212',2,503,'Madurai','computer,accessory','Black,White'),
    -> (106,'2026-01-15',2,'Bala','bala@gmail.com','Velachery','Chennai','600042','Tamil Nadu',201,'Laptop',55000,301,'Electronics',401,'Tech Supplies','9876543210',1,501,'Chennai','computer,electronics,work','Black,Silver'),
    -> (106,'2026-01-15',2,'Bala','bala@gmail.com','Velachery','Chennai','600042','Tamil Nadu',203,'Chair',2500,302,'Furniture',402,'Home Products','9876543211',1,502,'Coimbatore','furniture,office','Brown,Black');
Query OK, 12 rows affected (0.105 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- ======================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 2. 3NF / BCNF NORMALISED TABLES
Query OK, 0 rows affected (0.003 sec)

mysql> -- ======================================================
Query OK, 0 rows affected (0.003 sec)

mysql>
```

```
mysql> -- ======================================================
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> DROP VIEW IF EXISTS V_ORDER_HISTORY;
Query OK, 0 rows affected, 1 warning (0.043 sec)

mysql>
mysql> DROP TABLE IF EXISTS ORDER_ITEM;
Query OK, 0 rows affected, 1 warning (0.054 sec)

mysql> DROP TABLE IF EXISTS ORDERS;
Query OK, 0 rows affected, 1 warning (0.054 sec)

mysql> DROP TABLE IF EXISTS PRODUCT;
Query OK, 0 rows affected, 1 warning (0.056 sec)

mysql> DROP TABLE IF EXISTS WAREHOUSE;
Query OK, 0 rows affected, 1 warning (0.041 sec)

mysql> DROP TABLE IF EXISTS SHIP_ADDRESS;
Query OK, 0 rows affected, 1 warning (0.042 sec)

mysql> DROP TABLE IF EXISTS PIN_LOCATION;
Query OK, 0 rows affected, 1 warning (0.071 sec)

mysql> DROP TABLE IF EXISTS CUSTOMER;
Query OK, 0 rows affected, 1 warning (0.080 sec)

mysql> DROP TABLE IF EXISTS SUPPLIER;
Query OK, 0 rows affected, 1 warning (0.046 sec)

mysql> DROP TABLE IF EXISTS CATEGORY;
Query OK, 0 rows affected, 1 warning (0.042 sec)

mysql>
mysql> CREATE TABLE CUSTOMER (
    ->     CustomerID INT PRIMARY KEY,
```

```
mysql>
mysql> CREATE TABLE CUSTOMER (
    ->     CustomerID INT PRIMARY KEY,
    ->     CustomerName VARCHAR(100) NOT NULL,
    ->     CustomerEmail VARCHAR(120) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.541 sec)

mysql>
mysql> CREATE TABLE PIN_LOCATION (
    ->     ShipPIN VARCHAR(10) PRIMARY KEY,
    ->     ShipCity VARCHAR(80) NOT NULL,
    ->     ShipState VARCHAR(80) NOT NULL
    -> );
Query OK, 0 rows affected (0.423 sec)

mysql>
mysql> CREATE TABLE SHIP_ADDRESS (
    ->     AddressID INT PRIMARY KEY,
    ->     ShipStreet VARCHAR(150) NOT NULL,
    ->     ShipPIN VARCHAR(10) NOT NULL,
    ->     UNIQUE (ShipStreet, ShipPIN),
    ->     FOREIGN KEY (ShipPIN)
    ->         REFERENCES PIN_LOCATION(ShipPIN)
    ->         ON DELETE RESTRICT
    ->         ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.598 sec)

mysql>
mysql> CREATE TABLE SUPPLIER (
    ->     SupplierID INT PRIMARY KEY,
    ->     SupplierName VARCHAR(100) NOT NULL,
    ->     SupplierPhone VARCHAR(20) NOT NULL
    -> );
Query OK, 0 rows affected (0.351 sec)

mysql>
mysql> CREATE TABLE CATEGORY (
    ->     CategoryID INT PRIMARY KEY,
    ->     CategoryName VARCHAR(80) NOT NULL UNIQUE
    -> );
```

```
mysql>
mysql> CREATE TABLE PRODUCT (
    ->     ProductID INT PRIMARY KEY,
    ->     ProductName VARCHAR(100) NOT NULL,
    ->     UnitPrice DECIMAL(10,2) NOT NULL,
    ->     CategoryID INT NOT NULL,
    ->     SupplierID INT NOT NULL,
    ->     CHECK (UnitPrice >= 0),
    ->     FOREIGN KEY (CategoryID)
    ->         REFERENCES CATEGORY(CategoryID)
    ->         ON DELETE RESTRICT,
    ->     FOREIGN KEY (SupplierID)
    ->         REFERENCES SUPPLIER(SupplierID)
    ->         ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (0.638 sec)

mysql>
mysql> CREATE TABLE ORDERS (
    ->     OrderID INT PRIMARY KEY,
    ->     OrderDate DATE NOT NULL,
    ->     CustomerID INT NOT NULL,
    ->     AddressID INT NOT NULL,
    ->     FOREIGN KEY (CustomerID)
    ->         REFERENCES CUSTOMER(CustomerID)
    ->         ON DELETE RESTRICT,
    ->     FOREIGN KEY (AddressID)
    ->         REFERENCES SHIP_ADDRESS(AddressID)
    ->         ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (0.672 sec)

mysql>
mysql> -- The actual price paid is stored here.
Query OK, 0 rows affected (0.005 sec)

mysql> -- It will not change when PRODUCT.UnitPrice changes.
Query OK, 0 rows affected (0.004 sec)
```

```
mysql>
mysql> INSERT INTO SUPPLIER VALUES
    -> (401,'Tech Supplies','9876543210'),
    -> (402,'Home Products','9876543211'),
    -> (403,'Digital World','9876543212');
Query OK, 3 rows affected (0.060 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO CATEGORY VALUES
    -> (301,'Electronics'),
    -> (302,'Furniture');
Query OK, 2 rows affected (0.057 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO WAREHOUSE VALUES
    -> (501,'Chennai'),
    -> (502,'Coimbatore'),
    -> (503,'Madurai');
Query OK, 3 rows affected (0.061 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO PRODUCT VALUES
    -> (201,'Laptop',55000,301,401),
    -> (202,'Mouse',800,301,401),
    -> (203,'Chair',2500,302,402),
    -> (204,'Table',4500,302,402),
    -> (205,'Keyboard',1500,301,403);
Query OK, 5 rows affected (0.088 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>
```

```
mysql>
mysql> CREATE TABLE PRODUCT_TAG (
    ->     ProductID INT NOT NULL,
    ->     Tag VARCHAR(80) NOT NULL,
    ->     PRIMARY KEY (ProductID, Tag),
    ->     FOREIGN KEY (ProductID)
    ->         REFERENCES PRODUCT(ProductID)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.379 sec)

mysql>
mysql> CREATE TABLE PRODUCT_COLOUR (
    ->     ProductID INT NOT NULL,
    ->     Colour VARCHAR(50) NOT NULL,
    ->     PRIMARY KEY (ProductID, Colour),
    ->     FOREIGN KEY (ProductID)
    ->         REFERENCES PRODUCT(ProductID)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.429 sec)

mysql>
mysql> INSERT INTO PRODUCT_TAG VALUES
    -> (201,'computer'),
    -> (201,'electronics'),
    -> (201,'work'),
    -> (202,'computer'),
    -> (202,'accessory'),
    -> (203,'furniture'),
    -> (203,'office'),
    -> (204,'furniture'),
    -> (204,'office'),
    -> (205,'computer'),
    -> (205,'accessory');
Query OK, 11 rows affected (0.083 sec)
Records: 11  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE SUPPLIER_CATEGORY (
    ->     SupplierID INT NOT NULL,
    ->     CategoryID INT NOT NULL,
    ->     PRIMARY KEY (SupplierID, CategoryID),
    ->     FOREIGN KEY (SupplierID)
    ->         REFERENCES SUPPLIER(SupplierID),
    ->     FOREIGN KEY (CategoryID)
    ->         REFERENCES CATEGORY(CategoryID)
    -> );
Query OK, 0 rows affected (0.560 sec)

mysql>
mysql> CREATE TABLE CATEGORY_WAREHOUSE (
    ->     CategoryID INT NOT NULL,
    ->     WarehouseID INT NOT NULL,
    ->     PRIMARY KEY (CategoryID, WarehouseID),
    ->     FOREIGN KEY (CategoryID)
    ->         REFERENCES CATEGORY(CategoryID),
    ->     FOREIGN KEY (WarehouseID)
    ->         REFERENCES WAREHOUSE(WarehouseID)
    -> );
Query OK, 0 rows affected (0.680 sec)

mysql>
mysql> CREATE TABLE SUPPLIER_WAREHOUSE (
    ->     SupplierID INT NOT NULL,
    ->     WarehouseID INT NOT NULL,
    ->     PRIMARY KEY (SupplierID, WarehouseID),
    ->     FOREIGN KEY (SupplierID)
    ->         REFERENCES SUPPLIER(SupplierID),
    ->     FOREIGN KEY (WarehouseID)
    ->         REFERENCES WAREHOUSE(WarehouseID)
    -> );
Query OK, 0 rows affected (1.190 sec)

mysql>
mysql> INSERT INTO SUPPLIER_CATEGORY VALUES
    -> (401,301),
    -> (402,302),
    -> (403,301);
Query OK, 3 rows affected (0.090 sec)
```

```
mysql> CREATE VIEW V_ORDER_HISTORY AS
    -> SELECT
    ->     O.OrderID,
    ->     O.OrderDate,
    ->     C.CustomerID,
    ->     C.CustomerName,
    ->     C.CustomerEmail,
    ->     A.ShipStreet,
    ->     P.ShipCity,
    ->     P.ShipPIN,
    ->     P.ShipState,
    ->     I.ProductID,
    ->     PR.ProductName,
    ->     I.Quantity,
    ->     PR.UnitPrice AS CurrentPrice,
    ->     I.UnitPricePaid AS ActualPricePaid,
    ->     I.Quantity * I.UnitPricePaid AS LineTotal,
    ->     W.WarehouseCity,
    ->     S.SupplierName
    -> FROM ORDERS O
    -> JOIN CUSTOMER C
    ->     ON O.CustomerID = C.CustomerID
    -> JOIN SHIP_ADDRESS A
    ->     ON O.AddressID = A.AddressID
    -> JOIN PIN_LOCATION P
    ->     ON A.ShipPIN = P.ShipPIN
    -> JOIN ORDER_ITEM I
    ->     ON O.OrderID = I.OrderID
    -> JOIN PRODUCT PR
    ->     ON I.ProductID = PR.ProductID
    -> JOIN WAREHOUSE W
    ->     ON I.WarehouseID = W.WarehouseID
    -> JOIN SUPPLIER S
    ->     ON PR.SupplierID = S.SupplierID;
Query OK, 0 rows affected (0.452 sec)

mysql>
mysql>
mysql> -- ============================================================
Query OK, 0 rows affected (0.005 sec)
```

```
mysql> SELECT * FROM CUSTOMER;
+------------+--------------+------------------+
| CustomerID | CustomerName | CustomerEmail    |
+------------+--------------+------------------+
|          1 | Arun         | arun@gmail.com   |
|          2 | Bala         | bala@gmail.com   |
|          3 | Charan       | charan@gmail.com |
|          4 | Deepa        | deepa@gmail.com  |
+------------+--------------+------------------+
4 rows in set (0.008 sec)

mysql>
mysql> -- Display all products
Query OK, 0 rows affected (0.003 sec)

mysql> SELECT * FROM PRODUCT;
+-----------+-------------+-----------+------------+------------+
| ProductID | ProductName | UnitPrice | CategoryID | SupplierID |
+-----------+-------------+-----------+------------+------------+
|       201 | Laptop      |  55000.00 |        301 |        401 |
|       202 | Mouse       |    800.00 |        301 |        401 |
|       203 | Chair       |   2500.00 |        302 |        402 |
|       204 | Table       |   4500.00 |        302 |        402 |
|       205 | Keyboard    |   1500.00 |        301 |        403 |
+-----------+-------------+-----------+------------+------------+
5 rows in set (0.007 sec)

mysql>
mysql> -- Display all suppliers
Query OK, 0 rows affected (0.003 sec)

mysql> SELECT * FROM SUPPLIER;
+------------+--------------+---------------+
| SupplierID | SupplierName | SupplierPhone |
+------------+--------------+---------------+
|        401 | Tech Supplies | 9876543210    |
|        402 | Home Products | 9876543211    |
|        403 | Digital World | 9876543212    |
+------------+--------------+---------------+
3 rows in set (0.003 sec)

mysql>
```

```
mysql> SELECT * FROM ORDERS;
+---------+------------+------------+-----------+
| OrderID | OrderDate  | CustomerID | AddressID |
+---------+------------+------------+-----------+
|     101 | 2026-01-10 |          1 |         1 |
|     102 | 2026-01-11 |          2 |         2 |
|     103 | 2026-01-12 |          3 |         3 |
|     104 | 2026-01-13 |          4 |         4 |
|     105 | 2026-01-14 |          1 |         1 |
|     106 | 2026-01-15 |          2 |         2 |
+---------+------------+------------+-----------+
6 rows in set (0.002 sec)

mysql>
mysql> -- Display all order items
Query OK, 0 rows affected (0.001 sec)

mysql> SELECT * FROM ORDER_ITEM;
+---------+-----------+----------+-------------+---------------+
| OrderID | ProductID | Quantity | WarehouseID | UnitPricePaid |
+---------+-----------+----------+-------------+---------------+
|     101 |       201 |        1 |         501 |      55000.00 |
|     101 |       202 |        2 |         501 |        800.00 |
|     102 |       203 |        1 |         502 |       2500.00 |
|     102 |       204 |        1 |         502 |       4500.00 |
|     103 |       201 |        1 |         501 |      54000.00 |
|     103 |       205 |        1 |         503 |       1500.00 |
|     104 |       202 |        1 |         501 |        800.00 |
|     104 |       203 |        2 |         502 |       2500.00 |
|     105 |       204 |        1 |         502 |       4300.00 |
|     105 |       205 |        2 |         503 |       1500.00 |
|     106 |       201 |        1 |         501 |      53000.00 |
|     106 |       203 |        1 |         502 |       2500.00 |
+---------+-----------+----------+-------------+---------------+
12 rows in set (0.002 sec)

mysql>
mysql> -- Display complete order history
Query OK, 0 rows affected (0.001 sec)
```

```
mysql>
mysql> -- Display complete order history
Query OK, 0 rows affected (0.001 sec)

mysql> SELECT * FROM V_ORDER_HISTORY;
+---------+------------+------------+--------------+-----------------+-------------+-----------+---------+------------+-----------+-------------+----------+----------------+----------+---------------+---------------+
| OrderID | OrderDate  | CustomerID | CustomerName | CustomerEmail   | ShipStreet  | ShipCity  | ShipPIN | ShipState  | ProductID | ProductName | Quantity | CurrentPrice | ActualPricePaid | LineTotal | WarehouseCity | SupplierName  |
+---------+------------+------------+--------------+-----------------+-------------+-----------+---------+------------+-----------+-------------+----------+----------------+----------+---------------+---------------+
|     101 | 2026-01-10 |          1 | Arun         | arun@gmail.com  | Anna Nagar  | Chennai   | 600040  | Tamil Nadu |       201 | Laptop      |        1 |     55000.00 |        55000.00 |  55000.00 | Chennai       | Tech Supplies |
|     101 | 2026-01-10 |          1 | Arun         | arun@gmail.com  | Anna Nagar  | Chennai   | 600040  | Tamil Nadu |       202 | Mouse       |        2 |       800.00 |          800.00 |   1600.00 | Chennai       | Tech Supplies |
|     105 | 2026-01-14 |          1 | Arun         | arun@gmail.com  | Anna Nagar  | Chennai   | 600040  | Tamil Nadu |       204 | Table       |        1 |      4500.00 |         4300.00 |   4300.00 | Coimbatore    | Home Products |
|     105 | 2026-01-14 |          1 | Arun         | arun@gmail.com  | Anna Nagar  | Chennai   | 600040  | Tamil Nadu |       205 | Keyboard    |        2 |      1500.00 |         1500.00 |   3000.00 | Madurai       | Digital World |
|     104 | 2026-01-13 |          4 | Deepa        | deepa@gmail.com | Gandhipuram | Coimbatore | 641012 | Tamil Nadu |       202 | Mouse       |        1 |       800.00 |          800.00 |    800.00 | Chennai       | Tech Supplies |
|     104 | 2026-01-13 |          4 | Deepa        | deepa@gmail.com | Gandhipuram | Coimbatore | 641012 | Tamil Nadu |       203 | Chair       |        2 |      2500.00 |         2500.00 |   5000.00 | Coimbatore    | Home Products |
|     103 | 2026-01-12 |          3 | Charan       | charan@gmail.com | T Nagar    | Chennai   | 600017  | Tamil Nadu |       201 | Laptop      |        1 |     55000.00 |        54000.00 |  54000.00 | Chennai       | Tech Supplies |
|     103 | 2026-01-12 |          3 | Charan       | charan@gmail.com | T Nagar    | Chennai   | 600017  | Tamil Nadu |       205 | Keyboard    |        1 |      1500.00 |         1500.00 |   1500.00 | Madurai       | Digital World |
|     102 | 2026-01-11 |          2 | Bala         | bala@gmail.com  | Velachery   | Chennai   | 600042  | Tamil Nadu |       203 | Chair       |        1 |      2500.00 |         2500.00 |   2500.00 | Coimbatore    | Home Products |
|     102 | 2026-01-11 |          2 | Bala         | bala@gmail.com  | Velachery   | Chennai   | 600042  | Tamil Nadu |       204 | Table       |        1 |      4500.00 |         4500.00 |   4500.00 | Coimbatore    | Home Products |
|     106 | 2026-01-15 |          2 | Bala         | bala@gmail.com  | Velachery   | Chennai   | 600042  | Tamil Nadu |       201 | Laptop      |        1 |     55000.00 |        53000.00 |  53000.00 | Chennai       | Tech Supplies |
|     106 | 2026-01-15 |          2 | Bala         | bala@gmail.com  | Velachery   | Chennai   | 600042  | Tamil Nadu |       203 | Chair       |        1 |      2500.00 |         2500.00 |   2500.00 | Coimbatore    | Home Products |
+---------+------------+------------+--------------+-----------------+-------------+-----------+---------+------------+-----------+-------------+----------+----------------+----------+---------------+---------------+
12 rows in set (0.023 sec)

mysql>
mysql>
mysql> -- ==================================================
Query OK, 0 rows affected (0.002 sec)
```

```
mysql> SELECT
    ->     C.CustomerName,
    ->     O.OrderID,
    ->     O.OrderDate,
    ->     PR.ProductName,
    ->     I.Quantity,
    ->     I.UnitPricePaid,
    ->     I.Quantity * I.UnitPricePaid AS TotalAmount
    -> FROM CUSTOMER C
    -> JOIN ORDERS O
    ->     ON C.CustomerID = O.CustomerID
    -> JOIN ORDER_ITEM I
    ->     ON O.OrderID = I.OrderID
    -> JOIN PRODUCT PR
    ->     ON I.ProductID = PR.ProductID
    -> ORDER BY O.OrderDate;
+--------------+---------+------------+-------------+----------+---------------+-------------+
| CustomerName | OrderID | OrderDate  | ProductName | Quantity | UnitPricePaid | TotalAmount |
+--------------+---------+------------+-------------+----------+---------------+-------------+
| Arun         |     101 | 2026-01-10 | Laptop      |        1 |      55000.00 |    55000.00 |
| Arun         |     101 | 2026-01-10 | Mouse       |        2 |        800.00 |     1600.00 |
| Bala         |     102 | 2026-01-11 | Chair       |        1 |       2500.00 |     2500.00 |
| Bala         |     102 | 2026-01-11 | Table       |        1 |       4500.00 |     4500.00 |
| Charan       |     103 | 2026-01-12 | Laptop      |        1 |      54000.00 |    54000.00 |
| Charan       |     103 | 2026-01-12 | Keyboard    |        1 |       1500.00 |     1500.00 |
| Deepa        |     104 | 2026-01-13 | Mouse       |        1 |        800.00 |      800.00 |
| Deepa        |     104 | 2026-01-13 | Chair       |        2 |       2500.00 |     5000.00 |
| Arun         |     105 | 2026-01-14 | Table       |        1 |       4300.00 |     4300.00 |
| Arun         |     105 | 2026-01-14 | Keyboard    |        2 |       1500.00 |     3000.00 |
| Bala         |     106 | 2026-01-15 | Laptop      |        1 |      53000.00 |    53000.00 |
| Bala         |     106 | 2026-01-15 | Chair       |        1 |       2500.00 |     2500.00 |
+--------------+---------+------------+-------------+----------+---------------+-------------+
12 rows in set (0.009 sec)

mysql>
mysql>
mysql> -- ==================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 11. TOTAL ORDER VALUE
Query OK, 0 rows affected (0.002 sec)
```

```
mysql>
mysql> SELECT
    ->     P.ProductID,
    ->     P.ProductName,
    ->     P.UnitPrice,
    ->     C.CategoryName,
    ->     S.SupplierName,
    ->     S.SupplierPhone
    -> FROM PRODUCT P
    -> JOIN CATEGORY C
    ->     ON P.CategoryID = C.CategoryID
    -> JOIN SUPPLIER S
    ->     ON P.SupplierID = S.SupplierID;
+-----------+-------------+-----------+--------------+---------------+---------------+
| ProductID | ProductName | UnitPrice | CategoryName | SupplierName  | SupplierPhone |
+-----------+-------------+-----------+--------------+---------------+---------------+
|       201 | Laptop      |  55000.00 | Electronics  | Tech Supplies | 9876543210    |
|       202 | Mouse       |    800.00 | Electronics  | Tech Supplies | 9876543210    |
|       203 | Chair       |   2500.00 | Furniture    | Home Products | 9876543211    |
|       204 | Table       |   4500.00 | Furniture    | Home Products | 9876543211    |
|       205 | Keyboard    |   1500.00 | Electronics  | Digital World | 9876543212    |
+-----------+-------------+-----------+--------------+---------------+---------------+
5 rows in set (0.004 sec)

mysql>
mysql>
mysql> -- ========================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 13. CUSTOMERS WITH MORE THAN ONE ORDER
Query OK, 0 rows affected (0.001 sec)

mysql> -- ========================================================
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT
    ->     C.CustomerID,
    ->     C.CustomerName,
    ->     COUNT(O.OrderID) AS TotalOrders
    -> FROM CUSTOMER C
    -> JOIN ORDERS O
```

```
mysql>
mysql> SELECT
    ->     P.ProductName,
    ->     T.Tag,
    ->     C.Colour
    -> FROM PRODUCT P
    -> JOIN PRODUCT_TAG T
    ->     ON P.ProductID = T.ProductID
    -> JOIN PRODUCT_COLOUR C
    ->     ON P.ProductID = C.ProductID
    -> ORDER BY P.ProductID;
+-------------+-------------+--------+
| ProductName | Tag         | Colour |
+-------------+-------------+--------+
| Laptop      | computer    | Black  |
| Laptop      | electronics | Black  |
| Laptop      | work        | Black  |
| Laptop      | computer    | Silver |
| Laptop      | electronics | Silver |
| Laptop      | work        | Silver |
| Mouse       | accessory   | Black  |
| Mouse       | computer    | Black  |
| Mouse       | accessory   | White  |
| Mouse       | computer    | White  |
| Chair       | furniture   | Black  |
| Chair       | office      | Black  |
| Chair       | furniture   | Brown  |
| Chair       | office      | Brown  |
| Table       | furniture   | Brown  |
| Table       | office      | Brown  |
| Table       | furniture   | White  |
| Table       | office      | White  |
| Keyboard    | accessory   | Black  |
| Keyboard    | computer    | Black  |
| Keyboard    | accessory   | White  |
| Keyboard    | computer    | White  |
+-------------+-------------+--------+
22 rows in set (0.005 sec)
```

```
mysql>
mysql> DESCRIBE UNF_ORDER_LINE;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| OrderID       | int           | NO   | PRI | NULL    |       |
| OrderDate     | date          | NO   |     | NULL    |       |
| CustomerID    | int           | NO   |     | NULL    |       |
| CustomerName  | varchar(100)  | NO   |     | NULL    |       |
| CustomerEmail | varchar(120)  | NO   |     | NULL    |       |
| ShipStreet    | varchar(150)  | NO   |     | NULL    |       |
| ShipCity      | varchar(80)   | NO   |     | NULL    |       |
| ShipPIN       | varchar(10)   | NO   |     | NULL    |       |
| ShipState     | varchar(80)   | NO   |     | NULL    |       |
| ProductID     | int           | NO   | PRI | NULL    |       |
| ProductName   | varchar(100)  | NO   |     | NULL    |       |
| UnitPrice     | decimal(10,2) | NO   |     | NULL    |       |
| CategoryID    | int           | NO   |     | NULL    |       |
| CategoryName  | varchar(80)   | NO   |     | NULL    |       |
| SupplierID    | int           | NO   |     | NULL    |       |
| SupplierName  | varchar(100)  | NO   |     | NULL    |       |
| SupplierPhone | varchar(20)   | NO   |     | NULL    |       |
| Quantity      | int           | NO   |     | NULL    |       |
| WarehouseID   | int           | NO   |     | NULL    |       |
| WarehouseCity | varchar(80)   | NO   |     | NULL    |       |
| TagList       | varchar(500)  | YES  |     | NULL    |       |
| ColourList    | varchar(300)  | YES  |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
22 rows in set (0.015 sec)

mysql> DESCRIBE CUSTOMER;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| CustomerID    | int          | NO   | PRI | NULL    |       |
| CustomerName  | varchar(100) | NO   |     | NULL    |       |
| CustomerEmail | varchar(120) | NO   | UNI | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
3 rows in set (0.010 sec)
```

```
mysql> DESCRIBE ORDERS;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| OrderID    | int  | NO   | PRI | NULL    |       |
| OrderDate  | date | NO   | MUL | NULL    |       |
| CustomerID | int  | NO   | MUL | NULL    |       |
| AddressID  | int  | NO   | MUL | NULL    |       |
+------------+------+------+-----+---------+-------+
4 rows in set (0.009 sec)

mysql> DESCRIBE ORDER_ITEM;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| OrderID       | int           | NO   | PRI | NULL    |       |
| ProductID     | int           | NO   | PRI | NULL    |       |
| Quantity      | int           | NO   |     | NULL    |       |
| WarehouseID   | int           | NO   | MUL | NULL    |       |
| UnitPricePaid | decimal(10,2) | NO   |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
5 rows in set (0.009 sec)

mysql> DESCRIBE PRODUCT_TAG;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| ProductID | int         | NO   | PRI | NULL    |       |
| Tag       | varchar(80) | NO   | PRI | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
2 rows in set (0.009 sec)

mysql> DESCRIBE PRODUCT_COLOUR;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| ProductID | int         | NO   | PRI | NULL    |       |
| Colour    | varchar(50) | NO   | PRI | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
2 rows in set (0.009 sec)
```

```
mysql> SELECT 'UNF_ORDER_LINE' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM UNF_ORDER_LINE;
+---------------+-----------+
| TableName     | TotalRows |
+---------------+-----------+
| UNF_ORDER_LINE |        12 |
+---------------+-----------+
1 row in set (0.017 sec)

mysql>
mysql> SELECT 'CUSTOMER' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM CUSTOMER;
+-----------+-----------+
| TableName | TotalRows |
+-----------+-----------+
| CUSTOMER  |         4 |
+-----------+-----------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT 'PRODUCT' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM PRODUCT;
+-----------+-----------+
| TableName | TotalRows |
+-----------+-----------+
| PRODUCT   |         6 |
+-----------+-----------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT 'ORDERS' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM ORDERS;
+-----------+-----------+
| TableName | TotalRows |
+-----------+-----------+
| ORDERS    |         6 |
+-----------+-----------+
1 row in set (0.002 sec)
```

```
mysql> SELECT 'ORDER_ITEM' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM ORDER_ITEM;
+------------+-----------+
| TableName  | TotalRows |
+------------+-----------+
| ORDER_ITEM |        12 |
+------------+-----------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT 'PRODUCT_TAG' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM PRODUCT_TAG;
+-------------+-----------+
| TableName   | TotalRows |
+-------------+-----------+
| PRODUCT_TAG |        11 |
+-------------+-----------+
1 row in set (0.011 sec)

mysql>
mysql> SELECT 'PRODUCT_COLOUR' AS TableName,
    ->        COUNT(*) AS TotalRows
    -> FROM PRODUCT_COLOUR;
+----------------+-----------+
| TableName      | TotalRows |
+----------------+-----------+
| PRODUCT_COLOUR |        10 |
+----------------+-----------+
1 row in set (0.010 sec)

mysql>
mysql> -- ============================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- END OF SHOPNEST CO3 SQL SCRIPT
Query OK, 0 rows affected (0.001 sec)

mysql> -- ============================================================
Query OK, 0 rows affected (0.001 sec)
```